Name: Math 243: Calculus C

Exam 3 Solution Guide
26 April 2002

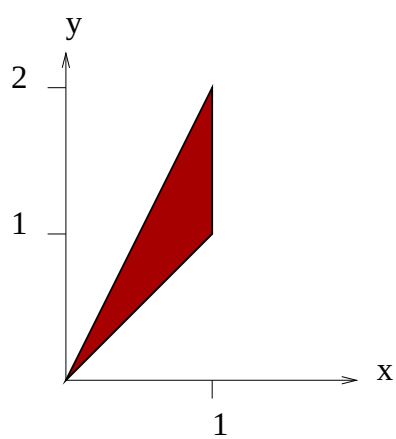
Always the beautiful answer who asks a more beautiful question.
–E. E. Cummings

Instructions: **Show all work to receive full or partial credit.** You may use a scientific calculator on this exam. All University rules and guidelines for student conduct are applicable.

1. [20 pts] Reverse the order of integration of the following double integral.

$$\int_0^1 \int_x^{2x} f(x,y)dydx.$$

The first step is to sketch the region of integration.



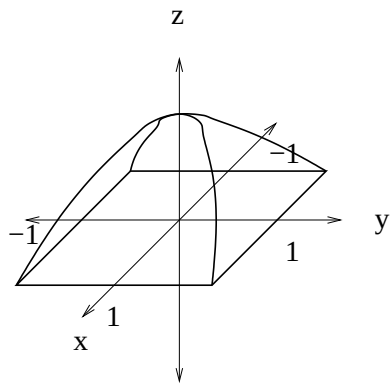
If we are going to integrate in y first, we must break the integral into two parts.

$$\int_0^1 \int_{\frac{y}{2}}^{y} f(x,y)dxdy + \int_1^2 \int_{\frac{y}{2}}^{1} f(x,y)dxdy.$$

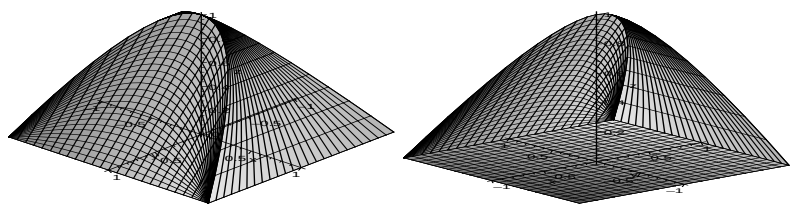
2. Consider the solid object bounded by the surfaces $z = 0$, $z + y^2 = 1$, $x + z = 1$ and $-x + z = 1$.

(a) [5 pts] Sketch the object labeling all axes carefully.

The toughest part of problems involving double and triple integrals is determining the proper region. After doodling around a little and roughly drawing each surface, you should see the intersection start to emerge. In the end, it will look like



which is not nearly as good as what I can do by hand. Or, if I ask Maple to sketch it,



(b) [15 pts] If the density of the object is

$$f(x, y, z) = 1 - z,$$

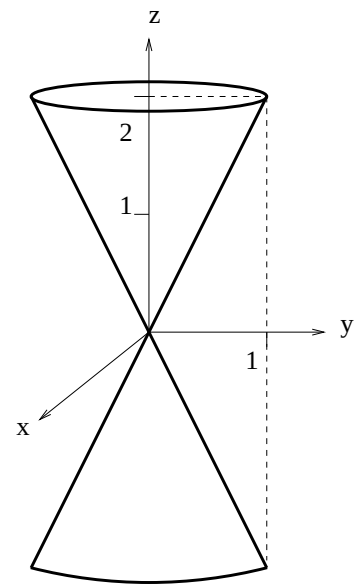
find the mass of the object.

Integrating the density over this region, we find that

$$\begin{aligned} m &= \int_{-1}^{1} \int_{0}^{1-y^2} \int_{z-1}^{1-z} (1-z)\,dx\,dz\,dy \\ &= \int_{-1}^{1} \int_{0}^{1-y^2} 2(1-z)^2\,dz\,dy \\ &= \int_{-1}^{1} \frac{2}{3}(y^6 - 1)\,dy \\ &= \frac{8}{7} \end{aligned}$$

This is one way to set up the triple integral. There are others, but this one is very simple.

3. [20 pts] Find the volume of the hour-glass.

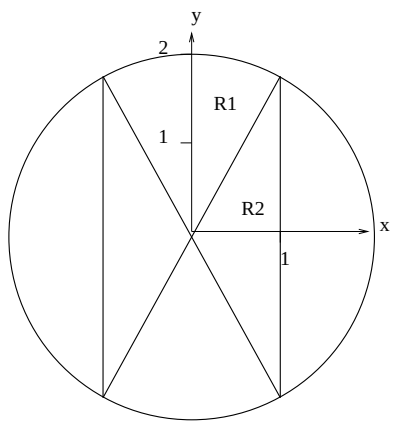


By symmetry, we can find the volume of one cone and then multiply by two. We see that the edge of the cone satisfies the equation $z = 2r$ if we use cylindrical coordinates. Thus, the total volume is

$$V = 2\int_0^{2\pi}\int_0^1\int_{2r}^2 r\,dz\,dr\,d\theta = \frac{4\pi}{3}.$$

4. [20 pts] Find the volume of the intersection of the cylinder $x^2 + y^2 \leq 1$ and the sphere $x^2 + y^2 + z^2 \leq 4$.

Either cylindrical or spherical coordinates will do the trick fairly directly for this problem. Both approaches will involve two pieces. Here, I present a solution using spherical coordinates. Let R_1 be the wedges at the top and bottom. Let R_2 be the cylinders with the wedges removed.



Again, we can use symmetry to compute the top half and multiply by two. The boundary between R_1 and R_2 can be found to be $\varphi = \frac{\pi}{6}$. For R_1, the limits of integration are fairly routine.

$$V_1 = 2\int_0^2 \int_0^{2\pi} \int_0^{\frac{\pi}{6}} \rho^2 \sin\varphi d\varphi d\theta d\rho = \frac{16\pi}{3}\left(2 - \sqrt{3}\right)$$

The second volume is a little more subtle because the outer bound on ρ corresponds to $r = 1$ in cylindrical coordinates. In other words, $\rho \sin\varphi = 1$, so $\rho = \csc\varphi$ is the outside boundary. Thus, the volume can be expressed by the integral

$$V_2 = 2\int_0^{2\pi} \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \int_0^{\csc\varphi} \rho^2 \sin\varphi d\rho d\varphi d\theta = \frac{4\pi}{3}\int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \csc^2\varphi d\varphi.$$

At this point, some students stopped because they could not remember that the antiderivative of $\csc^2\varphi$ is $\cot\varphi$. This is understandable, and getting this far is worth most of the points for the problem. To complete the problem, we see that

$$V_2 = \frac{4\pi}{3}\cot\varphi\Big|_{\frac{\pi}{6}}^{\frac{\pi}{2}} = \frac{4\pi\sqrt{3}}{3}.$$

Adding the two together, we arrive at the solution,

$$V = V_1 + V_2 = \frac{4\pi}{3}\left(8 - 3\sqrt{3}\right) = \frac{32\pi}{3} - 4\sqrt{3}\pi.$$

5. [20 pts] Suppose $f(x, y) = x^2 y$. Find

$$\int_C f(x, y) ds$$

where C is the line segment running from $(0, 0)$ to $(3, 4)$.

We can parametrize this line as

$$\mathbf{r}(s, t) = \langle 3, 4 \rangle t.$$

Then, we find that $ds = 5dt$. *Now, it is simply a matter of computing the path integral:*

$$\int_0^1 (3t)^2 (4t)(5dt) = 45.$$